Composer Classification from Symbolic (MIDI) Music Using Six Machine-Learning Models

UE24CS352A Machine Learning | Mini-Project Write-up | Based on Jain, Smit and Yngesjö, "Analysis and Classification of Symbolic Western Classical Music by Composer" (Stanford CS229, 2019)

1. Problem statement

Given the MIDI file of a Western classical piece, predict which composer wrote it. Composers of the same era write in very similar idioms: in a study cited by the reference paper, humans separated Mozart from Haydn only 57% of the time. We therefore need measurable style cues (harmony, melodic movement, texture), and we compare **six independent ML classifiers** on them and analyse which kinds of musical features carry the most composer information.

2. Dataset

MAESTRO v3.0.0 (Google Magenta): 1,276 MIDI recordings of piano-competition performances with composer metadata. The same piece is often played by several pianists, which would put identical music in both train and test, so we kept **one performance per (composer, title)** and removed 54 two-composer arrangements, leaving 785 pieces. Composers with fewer than 20 pieces were dropped, giving **725 pieces by 13 composers** (Chopin 101, Schubert 95, Bach 90, Liszt 90, Beethoven 86, Rachmaninoff 50, Debussy 40, Mozart 35, Schumann 31, Haydn 29, Scarlatti 29, Scriabin 26, Brahms 23). A stratified **60/20/20 split** gives 435 train, 145 dev and 145 test pieces. A synthetic set (360 fake-style MIDI files) was used only as a code smoke test; its ~100% accuracy has no musical meaning.

3. Approach

Each piece becomes a fixed-length vector of **186 music-theory features** in six families: pitch (64, including a key-normalised pitch-class histogram), dynamics (8), rhythm (34), melodic intervals (35), vertical intervals (19; dissonance ratio, parallel fifths/octaves) and texture (26; polyphony, instrument families). Features are standardised with training statistics only. The six models and their settings (from the paper where given):

Model	Core idea	Settings
Softmax regression	Linear score per composer, softmax probabilities	L2, C = 0.1
SVM (RBF)	Maximum-margin boundary; one-vs-one for 13 classes	C = 5.5, gamma = 1/186
LDA	Gaussian classes sharing one covariance	shrinkage = auto
k-NN	Distance-weighted vote of nearest pieces	k = 5
MLP	One hidden layer trained by SGD	150 ReLU units, lr 0.04, momentum 0.9, L2 1e-3
LightGBM	Boosted trees, each correcting earlier errors	700 rounds, lr 0.05, lambda 10, row/col subsample 0.6/0.5

The best model is chosen on the dev set and the test set is scored once. Metrics are accuracy and macro-averaged precision, recall and F1 (every composer counts equally).

4. Implementation overview

Python with *mido* (MIDI parsing), NumPy/pandas, scikit-learn, LightGBM and joblib (parallel extraction). **prepare_data.py** removes duplicate files, extracts features and writes the stratified split to features.csv; **train_tabular.py** trains and compares the six models and produces the analyses (confusion matrix, LightGBM gain by feature group, SVM accuracy versus PCA components, PCA scatter, pairwise composer classification, era classification per feature group); **predict.py** names the composer of a new MIDI file with the saved model.

5. Results

Model	Train acc.	Dev acc.	Test acc.	Macro-F1
LightGBM	1.000	0.834	0.869	0.800
SVM (RBF)	1.000	0.848	0.855	0.807
Softmax	0.998	0.793	0.855	0.790
MLP (1x150)	1.000	0.841	0.841	0.784
LDA (best on dev)	0.961	0.862	0.814	0.759
k-NN	1.000	0.697	0.759	0.660

Feature-group gain share (LightGBM): pitch 47%, melodic 27%, vertical 13%, rhythm 10%, dynamics 3%, texture 1%.

Era from one feature group (82 test pieces): all 0.988, pitch 0.976, melodic 0.963, rhythm 0.927, vertical 0.866, texture 0.732; always-romantic baseline 0.622.

Pairwise composer classification: mean 0.970.
SVM on PCA: dev 0.45 (2 components) to 0.86 (100).

6. Conclusions

- Gradient boosting (LightGBM) gave the highest test accuracy (0.869), as in the paper, with SVM and softmax within noise (0.855); k-NN was weakest (0.759), consistent with distance measures degrading in 186 dimensions.
- With 145 test pieces one piece is about 0.7 points, so 1-2 point differences between the top models are not statistically established.
- Pitch and melodic/vertical interval features dominate; rhythm and dynamics matter less per feature, and texture is almost uninformative because MAESTRO is piano-only. This agrees with the paper's finding that intervals matter more than rhythm.
- Limitations: 186 features instead of the paper's 1,494 (jSymbolic) and a different, piano-only dataset, so the paper's numbers are not reproduced; movements of one work may fall in different splits, so scores are probably optimistic; no cross-validation was done.